

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY



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Cardiovascular Disease Classification Using Machine Learning

1. Introduction

Machine learning can be used in healthcare to identify patterns in patient data and assist in disease prediction. In this assignment, a healthcare dataset was used to build a binary classification system for predicting the presence or absence of cardiovascular disease.

Three supervised machine learning algorithms were implemented and compared:

- Decision Tree
- Gaussian Naïve Bayes
- K-Nearest Neighbors (KNN)

The models were evaluated using Accuracy, Precision, Recall, and F1-score. The main objective was to determine which model performed best on the selected dataset.

2. Dataset Description

The Cardiovascular Disease Dataset was used for this assignment. The dataset contains information about 1,000 patients and originally contains 14 columns.

The target variable, target, is binary:

- 0 = Absence of heart disease
- 1 = Presence of heart disease

The original dataset contains patient information and several medical measurements, including age, gender, blood pressure, cholesterol, fasting blood sugar, chest pain, maximum heart rate, exercise-induced angina, and other cardiovascular-related measurements.

The original columns were:

patientid, age, gender, restingBP, serumcholesterol, fastingbloodsugar, chestpain, restingelectro, maxheartrate, exerciseangia, oldpeak, slope, noofmajorvessels, and target.

The dataset was examined using head(), shape, info(), and describe(). Initially, all 1,000 rows contained non-null values.

3. Data Preprocessing

Several preprocessing steps were performed before training the models.

First, the patientid column was removed because it is only an identifier and does not provide useful information for disease prediction.

The columns chestpain, restingelectro, and slope were also removed because they represent categorical information encoded using numerical values. The remaining input features were numerical.

The final input features were:

- age
- gender
- restingBP
- serumcholesterol
- fastingbloodsugar
- maxheartrate
- exerciseangia
- oldpeak
- noofmajorvessels

The target column was used as the output variable.

Although the dataset initially contained no conventional missing values, 53 records had a cholesterol value of 0. Since a cholesterol measurement of zero is not a realistic medical measurement, these values were treated as invalid/missing values. They were replaced with NaN and then imputed using the median of the valid cholesterol values. The median used was 326.0.

After preprocessing, the dataset contained 1,000 rows and 10 columns, consisting of 9 input features and 1 target variable. No missing values remained.

The data was then divided into training and testing sets using an 80:20 ratio:

- 800 samples for training
- 200 samples for testing

The split used `random_state=42` and stratification to maintain a similar target-class distribution in both sets.

Since KNN is a distance-based algorithm, feature scaling was performed using `StandardScaler`. The scaler was fitted only on the training data and then applied to both the training and testing data.

4. Methods

Three machine learning classification algorithms were used.

4.1 Decision Tree

A Decision Tree Classifier was trained using the original training features. A Decision Tree makes predictions by creating a series of decision rules based on the input features.

The model was initialized using `DecisionTreeClassifier(random_state=42)`.

4.2 Gaussian Naïve Bayes

A Gaussian Naïve Bayes classifier was used as the second model. It predicts the target class based on the probability distribution of the input features.

The model was initialized using `GaussianNB()` and trained using the original training data.

4.3 K-Nearest Neighbors (KNN)

KNN was used as the third classification model. It predicts the class of a data point based on the classes of its nearest neighboring data points.

The value of K was set to 5, meaning that the five nearest neighbors were considered for each prediction.

Because KNN relies on distance calculations, the standardized features were used for this model.

4.4 Evaluation Metrics

The models were evaluated using four metrics:

Accuracy: Measures the overall proportion of correctly classified samples.

Precision: Measures the proportion of predicted positive cases that were actually positive.

Recall: Measures the proportion of actual positive cases that were correctly identified.

F1-score: Provides a balance between Precision and Recall.

5. Results and Analysis

The performance of the three models on the 200 test samples is shown below.

Model	Accuracy	Precision	Recall	F1-score
Decision Tree	87.50%	88.24%	90.52%	89.36%
Naïve Bayes	87.00%	89.47%	87.93%	88.70%
KNN	88.50%	88.43%	92.24%	90.30%

The results show that KNN performed the best overall. It achieved the highest Accuracy (88.50%), Recall (92.24%), and F1-score (90.30%).

Naïve Bayes achieved the highest Precision (89.47%). This means that among the cases predicted as positive, Naïve Bayes had the highest proportion of truly positive cases. However, its Accuracy, Recall, and F1-score were slightly lower than those of KNN.

Decision Tree achieved an Accuracy of 87.50%, Precision of 88.24%, Recall of 90.52%, and F1-score of 89.36%. Its performance was relatively close to the other two models but was not the best overall.

KNN performed better overall in this experiment. One important reason is that its features were standardized before training. Since KNN uses distances between data points, scaling prevents features with larger numerical ranges from dominating the distance calculation. The high Recall of 92.24% also indicates that KNN was effective at identifying positive cases in the test set.

A bar chart was also created to visualize and compare the four evaluation metrics across the three models. The visualization showed that KNN generally achieved the strongest overall performance, while Naïve Bayes achieved the highest Precision.

6. Conclusion

In this assignment, three machine learning classification models—Decision Tree, Gaussian Naïve Bayes, and K-Nearest Neighbors—were implemented to classify cardiovascular disease.

The dataset was examined and preprocessed by removing irrelevant and categorical features, handling invalid cholesterol values, separating input features from the target variable, and dividing the data into training and testing sets. Feature scaling was applied for KNN because it is a distance-based algorithm.

Based on the experimental results, KNN was the best-performing model overall, achieving an Accuracy of 88.50%, Recall of 92.24%, and F1-score of 90.30%. Naïve Bayes achieved the highest Precision at 89.47%.

Therefore, based on the evaluation metrics used in this experiment, KNN provided the strongest overall performance for this cardiovascular disease classification dataset.